

Nicholas Eisenberg, Ph.D.

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Experience

Senior Data Scientist

June 2022 - Present

Nevada National Security Site

- Utilized YOLO v5 to develop an object-detection model in PyTorch that achieved a 80% mAP evaluation on an in-house thermal-image vehicle dataset for a R&D project.
- Designed an anomaly-detection model in PyTorch using an LSTM variational-autoencoder to detect component failure in a pulse-power x-ray machine. The model accurately predicted the location of a component failure with 87% accuracy and cut machine repair times by 50%.
- Applied the U-Net architecture with PyTorch to develop a deblurring model for a pulse-powered x-ray machine. The model was applied by the machine operators and reduced machine configuration time by 30%.
- Enhanced the scalability of models initially prototyped by our team of Analysts and Data Scientists for training on our internal high-performance computing (HPC) cluster. My role involved refactoring the existing code to enable efficient execution in a distributed and parallel setting.
- Mentored a summer intern in the development and execution of models on a HPC cluster, involving multi-node, multi-GPU setups and Slurm resource management.
- Developed and deployed a Django web interface with Docker Compose, enabling non-technical staff to add nodes and connections to the company's Neo4j graph database, without knowledge of the query syntax.

Personal Projects

iron.nvim Maintainer (GitHub link)

iron.nvim is an interactive REPL plugin for neovim written in lua. I have recently joined the developers of iron.nvim to help maintain the code base, add new features and respond to ongoing issues.

Reddit Comment Transformer Classifier (GitHub link)

I developed a transformer-based classifier to determine the subreddit origin for a reddit comment. I utilized PRAW to scrape 10,000 comments across 10 subreddits, where some of the subreddits shared a common theme and employed the BertTokenizer to integrate the comments into the model. I created a Flask application to interact with the model. The model achieved 73% accuracy among the 10 subreddits.

Technical Skills

Programming Languages: Python, SQL, Bash, Lua, Cypher, $\LaTeX$

Libraries: PyTorch, Pandas, Scikit-Learn, Scipy, Sqlalchemy, Matplotlib, Plotly, Django, Streamlit

Developer Tools: Git, AWS, Unix/Linux, Vim, Tmux, SLURM, Docker, Neo4j

Education

Auburn University - Ph.D.

June 2022

Research Area: Probability (Stochastic Processes and Partial Differential Equations)

Advisor: Le Chen

University of Nevada, Las Vegas - B.S./M.S.

December 2015 / January 2018

Major: Applied Mathematics

Publications

- 1: Eisenberg, N., Chen, L. Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics. Stoch PDE: Anal Comp (2022). <https://doi.org/10.1007/s40072-022-00258-6>
- 2: Eisenberg, N., Chen, L. Invariant Measures for the Nonlinear Stochastic Heat Equation with No Drift Term. J Theor Probab (2024). <https://doi.org/10.1007/s10959-023-01302-4>